Lab 02:
Operational Amplifier Applications

ECE 1212: Electronic Circuit Design Lab

2026 Fall

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Team Roles:

Activity	**Student Name**
Circuit Construction	Stefan Zhang
Data Collection	Nick Zhu
Data Analysis	Stefan Zhang; Nick Zhu
Report Writing	Stefan Zhang; Nick Zhu

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1) Objectives

The objectives of this experiment were to:

- build and test an instrumentation amplifier with nominal gain $K = -3$;
- investigate its input impedance and response to two input signals;
- study the relationship between input phase difference and output voltage;
- generate a ramp waveform using an op-amp integrator and relay;
- compare experimental results with Multisim simulations.

2) Equipment Used

Table 1: Equipment and components used in Lab 02

Item	**Quantity**
LM741 operational amplifier	3 or more
Dual DC power supply ($\pm 12\,\mathrm{V}$)	1 set
Function generator	1
Digital oscilloscope	1
Breadboard and connecting wires	1 set
$1\,\mathrm{k\Omega}$ resistors	as needed
$680\,\Omega$ resistor	1
Potentiometer	1
$10\,\mathrm{M\Omega}$ resistor	1
$100\,\mathrm{k\Omega}$ resistor	1
$0.10\,\mu\mathrm{F}$ capacitor	1
5-V-coil relay	1
Multisim simulation software	1

3) Part I: Instrumentation Amplifier

For the instrumentation amplifier,

$$V_o = -\frac{R_4}{R_3}\left(1 + \frac{2R_2}{R_1}\right)(V_2 - V_1).$$

With

$$R_1 = R_2 = R_3 = R_4 = 1\,\text{k}\Omega,$$

the nominal gain is

$$K = -3.$$

3.1) Part I(a): Gain Measurement

With V_1 grounded and a 2 kHz sinusoidal signal applied to V_2,

$$V_{2,pp} = 1.08\,\text{V}, \qquad V_{o,pp} = 3.16\,\text{V}.$$

Therefore,

$$|K| = \frac{3.16}{1.08} \approx 2.93.$$

This is close to the theoretical value of 3.



Figure 1: Measured input and output waveforms for Part I(a)

3.2) Part I(b): Input Impedance

A DC source was connected through

$$R_s = 10\,\text{M}\Omega.$$

The measured values were

$$V_s = 2.00\,\mathrm{V}, \qquad V_{in} = 0.828{,}76\,\mathrm{V}.$$

Using

$$Z_{in} = R_s \frac{V_{in}}{V_s - V_{in}},$$

the apparent input impedance is

$$Z_{in} = (10\,\mathrm{M\Omega}) \frac{0.82876}{2.00 - 0.82876} \approx 7.08\,\mathrm{M\Omega}.$$

3.3) Part I(c): Gain Adjustment

The original R_1 was replaced by a fixed resistor and a potentiometer:

$$R_1 = 680\,\Omega + 276\,\Omega = 956\,\Omega.$$

The theoretical gain becomes

$$K = -\left(1 + \frac{2(1000)}{956}\right) \approx -3.092.$$

The measured voltages were

$$V_{in,pp} = 1.08\,\mathrm{V}, \qquad V_{o,pp} = 3.32\,\mathrm{V},$$

giving

$$|K_{\mathrm{meas}}| = \frac{3.32}{1.08} \approx 3.074.$$

The measured gain is close to the theoretical value.

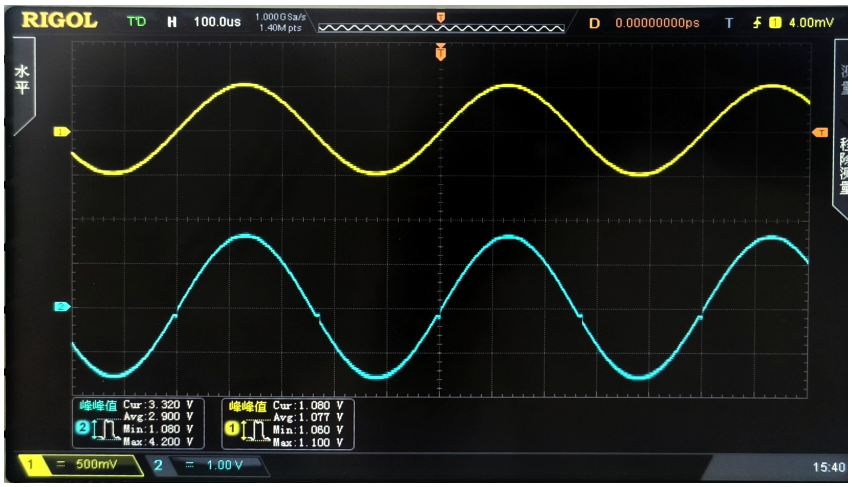


Figure 2: Measured waveform after gain adjustment

3.4) Part I(d): Two Input Signals

For Inputs 1,

$$V_1 = 1\,\mathrm{V}_{pp}, \qquad V_2 = 2\,\mathrm{V}_{pp},$$

with both signals at 1 kHz and in phase.

The theoretical output is

$$V_{o,pp} = 3(V_2 - V_1) = 3\,\mathrm{V}_{pp}.$$

The measured output was

$$V_{o,pp} \approx 3.08\,\mathrm{V}.$$



Figure 3: Measured output for Inputs 1

For Inputs 2,

$$V_1 = V_2 = 1\,\mathrm{V}_{pp}.$$

Ideally,

$$V_2 - V_1 = 0$$

and therefore

$$V_o = 0.$$

The measured residual output was approximately

$$V_{o,pp} = 140\,\mathrm{mV}.$$

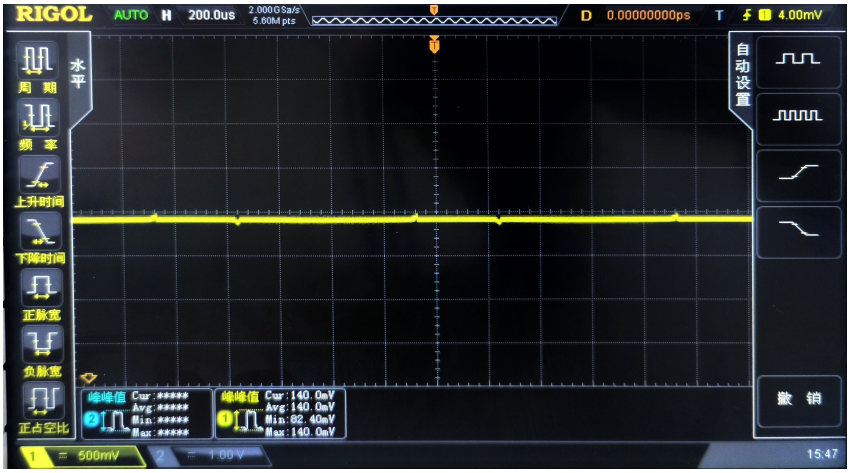


Figure 4: Measured output for Inputs 2

The small nonzero output is mainly caused by resistor mismatch, source mismatch, and non-ideal properties of the LM741.

3.5) Part I(e): Phase Difference

Both inputs were set to

$$V_1 = V_2 = 1\,\mathrm{V}_{pp}, \qquad f = 1\,\mathrm{kHz}.$$

For equal-amplitude sinusoidal inputs, the theoretical output is

$$V_{o,pp} = 6\left|\sin\left(\frac{\phi}{2}\right)\right|.$$

The results are shown below.

Table 2: Output voltage versus phase difference

Phase Difference	**Theoretical** $V_{o,pp}$	**Measured** $V_{o,pp}$
45°	2.30 V	2.38 V
90°	4.24 V	4.36 V
135°	5.54 V	5.68 V
180°	6.00 V	6.16 V

The output increases as the phase difference increases. Therefore, the magnitude of the phase difference can be estimated from

$$|\phi| = 2\sin^{-1}\left(\frac{V_{o,pp}}{6}\right).$$

The measured data agree well with the theoretical relationship.

4) Part II: Ramp Generation

The ramp generator used

$$R = 100\,\mathrm{k\Omega}, \qquad C = 0.10\,\mu\mathrm{F}, \qquad V_{PS} = 0.25\,\mathrm{V}.$$

When the normally-open relay contact is open, the circuit behaves as an inverting integrator:

$$\frac{dV_o}{dt} = -\frac{V_{PS}}{RC}.$$

Thus,

$$\frac{dV_o}{dt} = -\frac{0.25}{(100\,\mathrm{k\Omega})(0.10\,\mu\mathrm{F})} = -25\,\mathrm{V/s}.$$

The relay-control function generator was set to approximately

$$f = 3.33\,\mathrm{Hz}, \qquad V_{\mathrm{high}} = 6\,\mathrm{V}, \qquad V_{\mathrm{low}} = 0\,\mathrm{V}, \qquad D = 33.3\%.$$

The measured ramp waveform is shown below.



Figure 5: Measured ramp-generator output

The measured output change was approximately

$$V_{o,pp} = 5.16\,\mathrm{V},$$

which is close to the desired value.

The reset transition is not instantaneous because the electromechanical relay requires several milliseconds to switch.

5) Part III: Simulation

The instrumentation amplifier was simulated in Multisim using three LM741 op amps with $\pm 12\,\mathrm{V}$ power supplies.

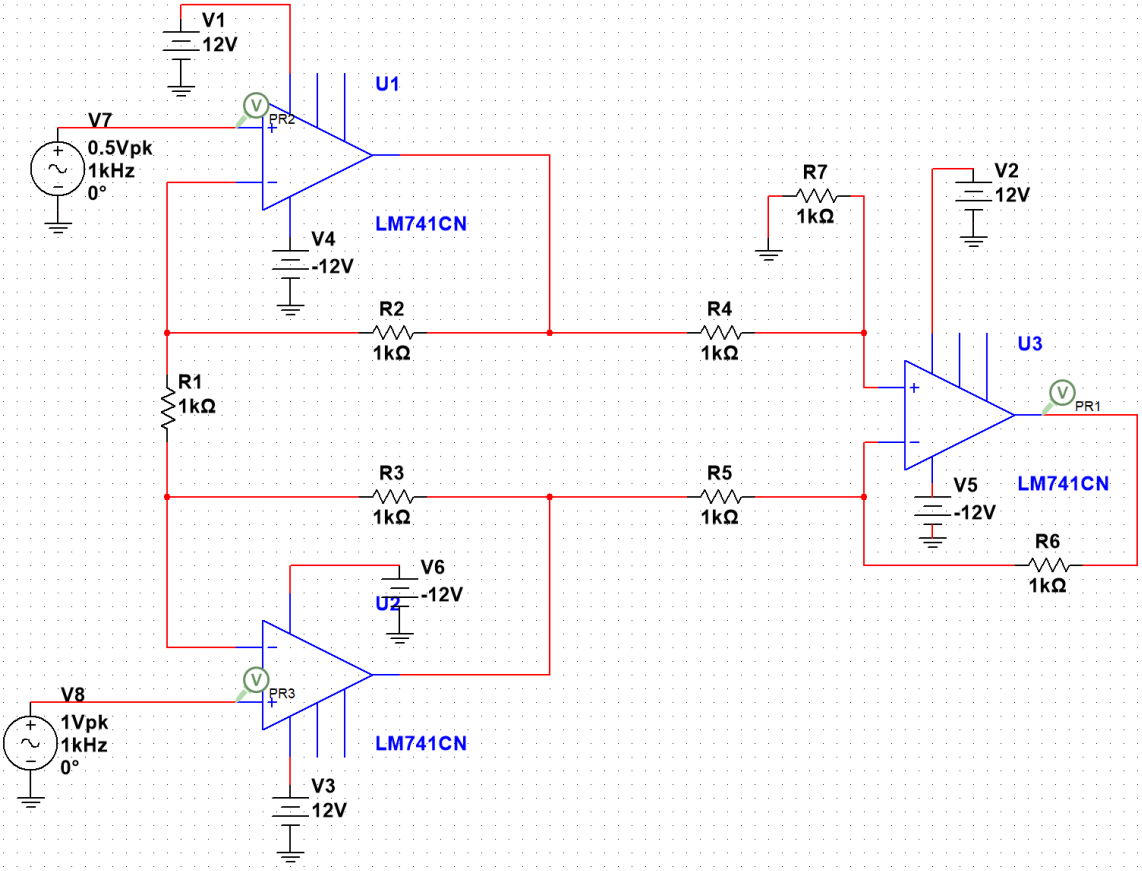


Figure 6: Multisim instrumentation-amplifier circuit

5.1) Inputs 1

The first simulation used

$$V_1 = 1\,\mathrm{V}_{pp}, \qquad V_2 = 2\,\mathrm{V}_{pp}, \qquad f = 1\,\mathrm{kHz}.$$

The expected output was

$$V_{o,pp} = 3\,\mathrm{V}.$$

The simulation produced approximately the same result.

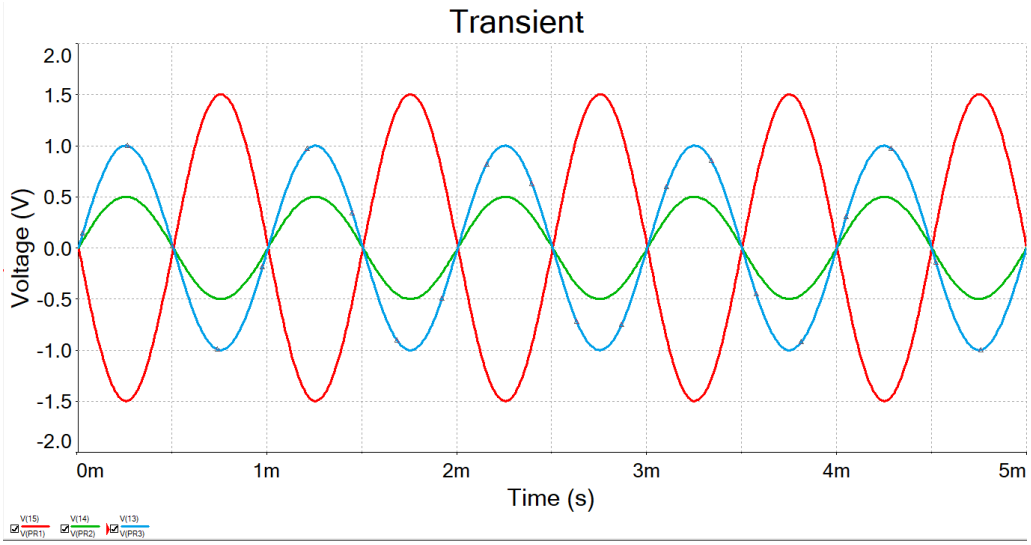


Figure 7: Simulation result for Inputs 1

5.2) Inputs 2

For

$$V_1 = V_2 = 1\,\mathrm{V}_{pp},$$

the theoretical output is

$$V_o = 0.$$

The simulation also produced approximately zero output.

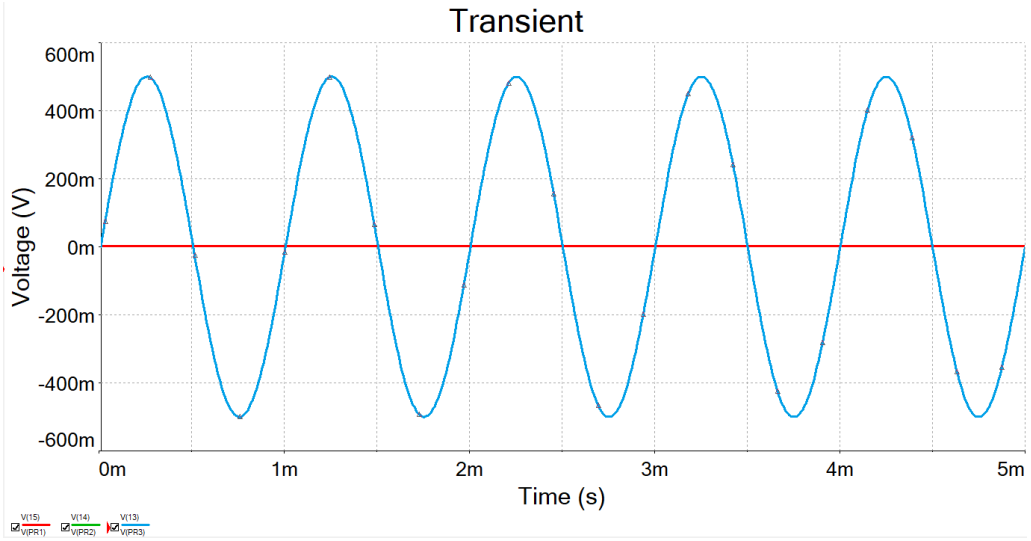


Figure 8: Simulation result for Inputs 2

Table 3: Comparison of experiment and simulation

Case	Experiment	Simulation
Inputs 1	$3.08\,\mathrm{V}_{pp}$	$\approx 3.00\,\mathrm{V}_{pp}$
Inputs 2	$0.14\,\mathrm{V}_{pp}$	$\approx 0\,\mathrm{V}$

The simulation agrees closely with the theoretical results. The experimental circuit shows small deviations because of component tolerances, op-amp non-idealities, and signal-source mismatch.

6) Conclusion

The instrumentation amplifier achieved a measured gain magnitude close to the designed value of 3. The output also followed the expected relationship with the amplitude and phase difference of the two input signals.

The ramp generator produced an approximately linear 5.16 V ramp, close to the required 5 V.

The Multisim results agreed with the theoretical behavior, while the small differences in the experimental results were consistent with practical component and measurement limitations.

References

[1] ECE 1212 Electronic Circuit Design Laboratory, *Experiment #2: Operational Amplifier Applications*, Fall 2026.

[2] A. S. Sedra and K. C. Smith, *Microelectronic Circuits*, 7th ed., Oxford University Press, 2015.